WU, GUANCHU

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RESEARCH INTERESTS

Quantum Information (Superconducting Quantum Computing, Quantum Neural Networks, Quantum Error Correction, Quantum Algorithms), AMO (Quantum Optics), Condensed Matter Theory, High Energy Physics

TECHNICAL SKILLS

Scientific Computing & Modeling: Machine Learning, Deep Learning, C++ and Python
Quantum & Theoretical Physics: Quantum Algorithms (e.g. QFT, Shor Code, Toric Code…)
Machine Learning & Data Analysis: NumPy and TensorFlow

EDUCATION

The Hong Kong University of Science and Technology HK SAR
Bachelor of Science in Physics (Class of 2027) Jun 2023 – Present

- **Cumulative GPA:** 3.365/4.3; Major GPA: 3.44/4.3
- **Awards and Honours:** University Scholarship (2023-24 Fall, 2023-24 Winter, 2023-24 Spring, 2023-24 Summer), Dean's List (2025-26 Fall)
- **Relevant Courses:** Real Analysis, Complex Analysis, Abstract Algebra, Differential Geometry, Quantum Mechanics, Methods of Experimental Physics, Thermodynamics and Statistical Physics, Programming with C++

Technical University of Munich München, Germany
Exchange Student Apr 2026 – Aug 2026

- **Courses:** Advanced Concepts of Quantum Computing, Introduction to Deep Learning, Quantum Information Theory, Theoretical Quantum Optics

RESEARCH EXPERIENCE

The Hong Kong University of Science and Technology HK SAR
Capstone Research Project, Supervised by Prof. HUANG, Shilin Aug 2026 – Present
Quantum Error Correction with Hyperbolic Surface Codes

- Constructed a finite {5,4} hyperbolic surface code from a group quotient; derived its X- and Z-check matrices from the lattice structure, verified that the stabilizers commute, and confirmed the [[60,8,4]] code parameters
- Used Stim to simulate four rounds of noisy syndrome extraction and PyMatching to compare standard and correlated matching decoders.
- Constructed a [[138,1,4]] ancilla code for a selected joint logical-Z measurement; verified the stabilizer relations and logical-operator mapping required to couple the ancilla to the data code.
- Plan to extend the simulation to include ancilla preparation, data–ancilla coupling and readout under noise; assess the fault tolerance of the complete measurement protocol and its qubit, gate and time requirements

Technical University of Munich München, Germany
Course Project, supervised by Prof. Stefan Filipp May 2026 – Aug 2026
Superconducting Quantum Computing

- Analytically and numerically studied the Hamiltonians of general superconducting circuits (e.g., transmon and coupled circuits) and a specific circuit from the experiment
- Applied various techniques (e.g., ladder operators and non-degenerate perturbation theory) to derive the energy levels, qubit frequency, anharmonicity, and eigenvectors
- Verified the analytical results by numerically diagonalizing the Hamiltonian in Python; interpreted the resulting mode structure with reference to the Hamiltonian

University of Electronic Science and Technology of China Chengdu, China
Research Project, Supervised by Prof. ZHANG, Xianggang Dec 2025 – March 2026
Zero-shot Control of a Simulated Robotic Arm based on OpenVLA

- Applies VLA model to realize zero-shot control of a simulated robot, aiming to realize autonomous decision-making from natural language instructions and visual observations
- Expands the application of the VLA model to a simulated block-sorting task in a college lab environment, and verifies its generalization capacity on unseen color combinations

- Designed evaluation tasks involving previously unseen color combinations and achieved a 64% success rate without task-specific fine-tuning, demonstrating the model's zero-shot generalization capability.

The Hong Kong University of Science and Technology HK SAR
Undergraduate Research Opportunities Program, supervised by Prof. YANG, Sen Jun 2025 – Sept 2025
UROP1000 - Quantum Computing based on Nitrogen-vacancy (NV) Center in Diamond

- Designed and carried out an optically detected magnetic resonance (ODMR) experiment to measure the spin resonance spectrum of the nitrogen-vacancy (NV) center in diamond
- Derived principles of quantum computing, including Rabi oscillation, optical pumping, rotating wave approximation (RWA), and optically detected magnetic resonance (ODMR); analyzed characteristics of the NV center
- Developed the camera driver for an Intensified Charge-Coupled Device (ICCD), specifically fan control and gate mode functions; integrated external pulse generator, microwave generator, and the ICCD camera into a unified control program; programmed to control pulse sequences and microwave and trigger the ICCD acquisition
- Captured the photon generated by the NV center, and resolved the energy level splitting of the NV center

West China Second University Hospital, Sichuan University Chengdu, China
Research Project, supervised by Prof. LIAO, Shujuan Dec 2024 – Feb 2025
Study on the Influencing Factors and Prediction Models of Delayed Onset of Lactation in Postpartum Women with Maternal-Infant Separation

- Performed data cleaning, missingness analysis, and multiple imputations on data from 701 parturient women
- Utilized Python to build and tune machine learning models, including RF, LASSO, binary logistic regression, SVM, XGBoost, and Gradient Boosting Machine
- Assisted in generating discrimination (AUC), calibration plots, and DCA curves; compared model performance across all metrics
- Successfully constructed and internally/externally validated five DOL prediction models (the optimal model Logistic regression reaching AUC = 0.885 and calibration slop = 0.996)

West China Second University Hospital, Sichuan University Chengdu, China
Research Project, supervised by Prof. CAI, Xiaotang June 2024 –August 2024
Differential Effects of Robot Assisted Gait Training in Children with Cerebral Palsy

- Assisted 37 children with cerebral palsy in robot-assisted gait training sessions, ensuring their safety and comfort throughout the intervention
- Collected primary and secondary outcome data, including resting heart rate, heart rate variability, Gross Motor Function Measure (GMFM) scores, and WeeFIM functional independence ratings at scheduled assessment time points
- Organized, entered, and cleaned the longitudinal dataset; verified data accuracy, and checked for outliers

ADDITIONAL INFORMATION

Interests: Basketball
Language Skills: Mandarin Chinese (native), English (fluent)